Project Design Phase-I
Proposed Solution Template

Date	2 March 2026
Team Members Name	Aditi Parashar , Adarsh Singh , Aditya
Project Name	Project - **Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau**
Maximum Marks	2 Marks

Proposed Solution Template:

Project Details:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Real estate agents, home buyers, and property investors struggle to make informed decisions due to the lack of clear, visual insights from large housing datasets. With over 21,609 housing records spanning prices from $50K to over $1.1M, manually analyzing variables such as sale price, house age, number of bedrooms, bathrooms, floors, basement area, and renovation status is time-consuming and ineffective. The absence of an interactive data visualization tool makes it difficult to identify market trends, understand price distributions, and compare property features efficiently.
2.	Idea / Solution description	An interactive multi-view Tableau dashboard was developed to analyse 21,609 housing records and transform raw data into clear, actionable visual insights. The solution includes KPI summary cards displaying an average sale price of $511,619, total basement area of 38.6M sqft, and a total property count of 21,609. A bubble chart visualizes market concentration across price ranges, a bar chart shows the distribution of house sale prices from $50K to $1.1M, a grouped bar chart compares bedrooms, bathrooms, and floors across different house age groups, and a pie chart breaks down house age distribution by renovation status among 204 renovated properties.
3.	Novelty / Uniqueness	Unlike static spreadsheets or generic listing platforms, this solution provides a unified, interactive Tableau dashboard that combines multiple dimensions of housing data in one place. It uniquely allows users to simultaneously explore price distributions, structural features, house age, and renovation status through visually rich and easy-to-

		understand charts. The use of a bubble chart for market concentration, combined with KPI cards and a story-driven dashboard layout, offers a comprehensive and non-technical way to interpret complex housing market data that is not available in any existing single tool.
4.	Social Impact / Customer Satisfaction	This solution empowers first-time home buyers to make confident, data-backed purchasing decisions within their budget range. Real estate agents benefit from improved pricing strategies based on clear market concentration insights. Property investors gain the ability to identify high-opportunity price segments quickly. By making complex housing data accessible to non-technical users, the solution reduces financial risk, saves decision-making time, and promotes fairer, more transparent property transactions — positively impacting the broader housing market community.
5.	Business Model (Revenue Model)	The solution can be offered as a subscription-based analytics service for real estate agencies, providing monthly access to updated Tableau dashboards with fresh housing market data. Individual reports and insights can be sold to property investors and buyers on a pay-per-report basis. Partnerships with real estate platforms and agencies can generate licensing revenue. Additionally, consulting services can be offered to agencies requiring customized dashboards tailored to specific regions or property types, creating a diversified and scalable revenue stream.
6.	Scalability of the Solution	The Tableau dashboard solution is highly scalable and can be expanded to include additional housing datasets from multiple cities, regions, or countries. New variables such as neighbourhood crime rates, proximity to schools, or property tax data can be integrated to enrich analysis. The solution can be published on Tableau Server or Tableau Online to support multiple simultaneous users across large organizations. As data volume grows, the underlying dataset can be connected to live databases or cloud storage solutions, ensuring the dashboard remains up to date and continues to deliver real-time housing market insights at scale.